

Annotating Barches & Slafish,
with least effort.

Annotate B's by finding the latest
visible ones & propagate backwards
For SF, they move, so how to?

SF: ¹⁰ amount of img per 15 min.

If I could see video it would be nice,
or is I'll look @ pictures to see when
the B/SF become invisible

08/08 - 4/gae

08/01 - "

07/30 - "

07/22 - visible

07/22 Algae starts, but moving?

· good time ~ 11am bc of light

utils

network drive
as a Fibreshare
network attached
storage

SAMBA protocol

Tools / timelapse generator - gui.py

INPUT
Lotus img

Rig is setup
demo_all / default = same settings
20p Auto exp 20% lens exposure

OUTPUT:
OUTPUT.mp4

Rig 1
20 Auto exp

date ...
Light config demo All

Preview match count to see which

10img for capturing every 15min

Mount Vop - data -> folder pointing to seq

LOTUS PTO proposal aims to develop and validate innovative laser-structured surfaces, mimicking natural surface

structures, for improved biofouling resistance and tribological performance through both simulations and experimental

testing, advancing from proof of concept (TRL3) to ocean environment validation (TRL5), dedicated mainly to critical

components of wave and tidal energy converts, in particular the PTO. It is a very interesting proposal with only a few

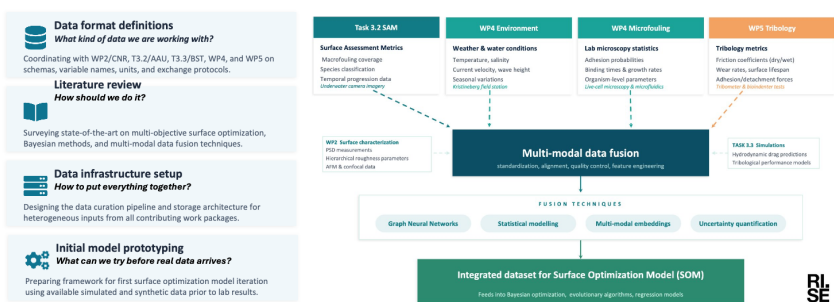
issues. a very well prepared document, the workplan is convincing. a unique problem is targeted.

The main objective of LOTUS-PTO is to utilize laser-structured surface technology to improve the performance and reliability of future ocean renewable energy technologies.

LOTUS-PTO will crosscut offshore renewable technologies with material design that reduce costs through a nontoxic strategy to improve biofouling management and through optimized tribological performance. Additionally, LOTUS-PTO will take knowledge on the technology of creating biomimicry inspired surface structures using laser power from an experimental proof of concept level (TRL3) to where it is validated in an ocean environment (TRL5).



T3.4 AI model for surface optimization

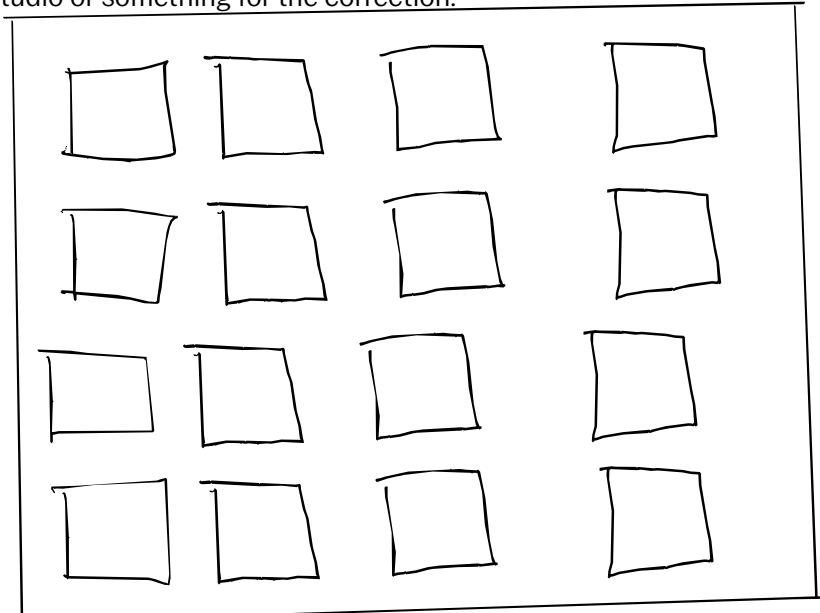


notes after seeing the presentations:

- trying to build a pipeline for first recognising the species, and then monitoring growth through the time.
- I need to figure out a way to annotate the data, malte suggested annotating the final images, with the big barnacles, and propagating back to when they are small. a part from how disgusting they are, I can manage.
- starfish are more tricky, as they move. could figure out how fast they move and annotate once every so often/one image per capture time, and propagate to the rest.
- problem for barnacles, a part from disgusting:
 - algae cover the images.
 - they might grow through time, so we need to figure out how to see when they start growing. so I start at the end, annotate a set of images, and see how far back I can go to monitor that they are still there and they are not new: if I annotate one now, but it is only two weeks old, if I propagate it back to three weeks back, it is not there
 - thus: we can crop out the images to make it more digestible (each square) from the latest visible set of images. annotate those with point annotation, and then sample every week maybe, to see if the points match. if they don't, then we find within the time range from when they were found to when they weren't and set a marker for that, so we know when they start growing. we can sample from within the week half way through and then half again, to see when they appear more accurately.
 - problem: confirmation bias. if I see the point, but can't see anything behind, then I might either think that they are there or aren't. design a good plan for that: eg. make a few "from scratch" annotations, and match them every once in a while, so that we are sure?
- starfish: they should move. but how much? can we run the most basic starfish detector? eg. countgd? can I use countgd for the barnacles? please?

TEST OUT WITH COUNTGD once I have a powerful computer. it could make annotations much faster?

if I did that, it would be with a higher threshold, and then migrating the data to label studio or something for the correction.



take the individual plates, count on there for closeup-ness, and then propagate back to a few weeks. how long does a barnacle take to grow.

so having visible/ semi transparent points on the pictures could be helpful to mitigate confirmation bias.